

1. **ISCED Code:** 0613
2. **Course Code:** CSE-2321
3. **Course Title:** Data Structures
4. **Type:** Core, Engineering
5. **Semester:** 3rd
6. **Credit Hours:** 3
7. **Contact Hours:** 3 lecture hours per week
8. **CIE Marks :** 50
9. **SEE Marks:** 50
10. **Total marks:** 100
11. **Prerequisite:** CSE-1121 (Computer Programming 1)
12. **Co-requisite:** CSE-2322 (Data structure Lab)
13. **Academic Session:** Spring 2024

14. Instructor's Details

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15. Course Rationale / Summary: This course is a continuation to the introduction to computer science and is a study of the different types of data structures, their design, implementation, efficiency and effective use in solving problems. It introduces students to new types of data structures such as arrays, linked lists, trees, graphs, heaps, stacks and queues. Students will also learn how to design algorithms for each new data structure studied, create and perform simple operations such as insertion, deletion, merging, sorting, and traversing on data structures. It describes and implements common algorithms for working with advanced data structures and recognizes which data structure is the best to use to solve a particular problem. To take this course, students should be able to program in a standard programming language preferably in C/C++. Some mathematical maturity will also be helpful for the students.

16. Course Objective: Upon completion of the course, students will be able to:

1. Impart a thorough understanding of linear data structures such as linked list, arrays, stacks, queues and their applications.
2. Learn a thorough understanding of non-linear data structures such as trees, graphs and their applications..
3. Familiarize with various sorting, searching and hashing techniques and their performance comparison.
4. Design and analyze recursive algorithms in data structures.

17. Course Outcomes (COs):

Upon successful completion of this course, students will be able to:

#	CO Description	Weightage
1.	Describe the operations of different linear and non-linear data structures along with the time and space complexity	50%
2.	Apply appropriate data structures to solve different problems	30%
3.	Analyze the performance of different data structures to solve problems	20%

18. Mapping of CO-PO-WK-WP-WA:

#	COs	POs	DL	KP	EP	EA	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CO1	Describe the operations of different linear and non-linear data structures along with the time and space complexity	PLO 1	C2	K3	-	-	Lecture CT, Quiz	CT, Quiz Exam
CO2	Apply appropriate data structures to solve different problems	PLO 1	C3	K3	-	-	Lecture, Assignment	Assignment Exam
CO3	Analyze the performance of different data structures to solve problems	PLO 2	C4	K3	-	-	Lecture, Assignment	Assignment Exam
Note: DL: Domain/level of learning taxonomy, KP: Knowledge Profile, EP: Attribute of Complex Engineering Problems, EA: Attribute of Complex Engineering Activities Learning Domains (C: Cognitive, A: Affective, P: Psychomotor)								

19. Course Content

#	Content	Duration	CLOs
Mid-Term (30 Marks)			
1	Introduction: Elementary Data organization, Information; Data types; Data Structure, Data Structure operations; Algorithm; Time-Space tradeoff of Algorithms. Mathematical notation & Functions; Algorithmic Notation; Control structures; Sub-algorithms. String; String operations; Pattern matching algorithms	06	CLO1 CLO2
2	Array: Linear Array (LA) & its representation in memory; Traversing LA, Insertion & Deletion in LA, Bubble Sort, Linear Search & binary Search. 2D Array & its representation in memory; Matrices; Algebra of matrices; sparse matrices	06	CLO1 CLO2
3	Linked list - Linked list & its representation in memory; Traversing, Searching, Insertion & Deletion operation on Linked list; Header linked lists; two way lists.	06	CLO1 CLO2

Final Exam: 50 Marks Group-A (20 Marks)			
4	Stack: its representation & applications; PUSH and POP operation on stack. Polish Notation, reverse polish notation; Evaluation of a postfix expression; Transforming infix expression into postfix expression.	03	CLO1 CLO2
5	Queue – its representation; Insertion & deletion in Queue; Deques; Priority Queues. Recursion [Factorial function, Fibonacci sequence, Ackermann function, Towers of Hanoi]	06	CLO1 CLO2
Group-B (30 Marks)			
6	Complexity of algorithms, Rate of growth: Big O, Ω and Θ notations; Complexity of Linear Search, Binary search & Bubble sort algorithm. Sorting - Insertion sort, selection sort, quick sort, merge sort; Searching & data modification; Hashing: Hash function, collision resolution	06	CLO2 CLO3
7	Tree - Tree terminology; representation of binary trees in memory; Traversing binary tree; Binary search tree; Insertion & deletion on binary search tree; Heap; Insertion & deletion on heap; Heapsort; B trees; General tree; Balanced binary search tree (AVL tree, red-black tree)	06	CLO1 CLO2
8	Graph – graph terminology; representation of graphs – adjacency matrix, path matrix, adjacency list; Traversing a graph – BFS & DFS	06	CLO1 CLO2

20. Weekly Activity Plan:

Week	Topic	TLS	AS	CLOs
Week 1	Elementary Data organization, Information; Data types; Data Structure, Data Structure operations; Algorithm; Time-Space tradeoff of Algorithms.	Lecture	CT, Quiz	CLO1
Week 2	Mathematical notation & Functions; Algorithmic Notation; Control structures; Sub-algorithms. String ; String operations; Pattern matching algorithms	Lecture, Discussion	Quiz, Assignment	CLO1
Week 3	Linear Array & its representation in memory; Traversing LA, Insertion & Deletion in LA, Bubble Sort, Linear Search & binary Search.	Lecture, Discussion	Class work, Quiz, CT	CLO1
Week 4	2D Array & its representation in memory; Matrices; Algebra of matrices; sparse matrices	Lecture, Discussion	CT, Quiz, Assignment	CLO2
Week 5	Linked list - Linked list & its representation in memory; Traversing, Searching	Lecture, Lab work, Assignment	Assignment, CT	CLO1
Week 6	Insertion & Deletion operation on Linked list; Header linked lists; two way lists.	Lecture, Lab work, Assignment	Assignment, CT	CLO1

Week	Topic	TLS	AS	CLOs
Week 7	Stack: its representation & applications; PUSH and POP operation on stack. Polish Notation, reverse polish notation; Evaluation of a postfix expression; Transforming infix expression into postfix expression.	Lecture, Discussion	CT, Assignment	CLO1 CLO2
Week 8	Queue – its representation; Insertion & deletion in Queue; Deques; Priority Queues.	Lecture, Lab work, Assignment	Assignment, Quiz	CLO1
Week 9	Recursion [Factorial function, Fibonacci sequence, Ackermann function, Towers of Hanoi]	Lecture, Lab work, Assignment	Assignment, Quiz	CLO2
Week 10	Complexity of algorithms, Rate of growth: Big O, Ω and Θ notations; Complexity of Linear Search, Binary search & Bubble sort algorithm	Lecture, Lab work, Assignment	Assignment, Quiz	CLO3
Week 11	Sorting - Insertion sort, selection sort, quick sort, merge sort; Searching & data modification; Hashing: Hash function, collision resolution	Lecture, Lab work, Assignment	CT, Assignment	CLO2 CLO3
Week 12	Tree - Tree terminology; representation of binary trees in memory; Traversing binary tree; Binary search tree; Insertion & deletion on binary search tree;	Lecture, Lab work, Assignment	Assignment, CT	CLO2
Week 13	Heap; Insertion & deletion on heap; Heapsort; B trees; General tree; Balanced binary search tree (AVL tree, red-black tree)	Lecture, Discussion	Class work, Assignment	CLO2
Week 14	Graph – graph terminology; representation of graphs – adjacency matrix, path matrix,	Lecture, Lab work, Assignment	Assignment, CT	CLO2
Week 15	Adjacency list;, Traversing a graph – BFS & DFS	Lecture, Lab work, Assignment	Assignment, CT	CLO2

Part C

21. Assessment Strategy:

The assessment and evaluation strategies for the course are given as follows:

Assessment Strategies	Description
Class Participation:	Students' individual in-class responses, attention, and sense of discipline, morality will be judged on the basis of 10 (ten) marks.
Class Test/Quiz:	Students will sit f 2 (Two) class test/quiz during the semester. One test/quiz will be taken before midterm and another after mid term. Class test/quiz marks will be assessed in 4 (four). Best of one class test/quiz will be taken. Students are strongly recommended not to miss any test.
Group Work:	The students will have to form groups consisting of a maximum of 4 members. There will be 1 assignment consisting of 3 (Three) marks. The topics or case studies will be given as assignments in groups during the class which they have to prepare at home and will submit on or before the due date.

Assessment Strategies	Description
Assignment:	There will be 1 assignment consisting of 3 (Three) marks. Some problems will be given as assignments during the class which they have to prepare at home and will submit on or before the due date.
Oral Presentation:	
Viva-vocé	
Midterm Exam:	Midterm exam will be held according to the Academic Calendar published by the university. Midterm assessment marks will be 30 (thirty).
Final Exam:	Final exams will be held according to the Academic Calendar published by the university. Final assessment marks will be 50 (fifty). Course contents learnt before Midterm examination will not be included in the syllabus of Final Examination.
Make-up Procedure:	No late submission and/or make-up assignment/presentation/quiz will be allowed without prior permission and adequate and reasonable proof of absence.

22. Marks Distribution:

Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50 marks)
Cognitive learning	Affective learning	Attendance Marks (10)	Assignment/ Class Test (10)	Mid-term (30)	Written Exam (50)
Remember					
Understand			4	16	25
Apply			3	9	15
Analyze			3	5	10
Evaluation					
Create					
	Responding	10			
Total allocated marks		10	10	30	50

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination

Delivery methods & activities: Lecture, White Board Writing, Questions and Answers, Discussions PowerPoint Presentation,

Assessment tools: Class Attendance, Class test/Quiz, Assignment on problem solution, Mid-Term & Final Exam, Viva

23. Grading Policy: As per IIUC grading policy

24. Code of Conduct:

- It is mandatory for all the students to participate in the class regularly and maintain proper discipline in the class.
- If a student fails to attend any class test, term exams, or final examination, he/she will get a zero in that class test, term, or final examination.
- Adopting unfair means in the exams will be considered as a serious crime and the student shall be placed to the university disciplinary committee.

- d. All the assignments, class test and exam copies should be neat and clear and demonstrate professionalism.
- e. No student is allowed to duplicate other student's work directly or with minor changes.
- f. Plagiarism is strictly restricted. One needs to provide a reference while using someone else's words, ideas, or research in assignments/exams.

Part D

25. Learning Materials:

Text Books:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1	Seymour Lipschutz	Data Structures	Special Indian Edition	Tata McGraw-Hill	2014	ISBN-13: 978-0-07-060168-0

Reference Books:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1.	Y. Langsam, Augenstein, A. M. Tanenbaum	Data Structures Using C and C++	2 nd Edition	Prentice Hall India	2014	ISBN:13 978-0387202778
2.	Edward M. Reingold, Wilfred J. Hansen	Data Structures	1 st Edition	CBS Publishers and Distributors	1983	ISBN-13: 978-0316739511
3.	Robert Sedgewick	Algorithms in C	3 rd	Pearson Education, Inc	2001	ISBN-13: 978-0321573513
4.	D. Samanta	Classic Data Structures	2 nd	Prentice Hall of India	2003	ISBN-10: 8120318749